

A Dynamical Field Model of Music-Induced Affect: Beyond the Valence–Arousal Circumplex

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Abstract

The dominant framework for modelling music-induced affect is Russell’s valence–arousal circumplex: a static two-dimensional map on which emotional states are located as points. The framework describes *where* a listener is emotionally, but not *how* they move there, what forces act on them, what traps them, or what allows escape. We present a dynamical field model of music-induced affect in which emotional state is a continuous vector $\mathbf{e}(t) \in \mathbb{R}^{16}$, governed by a Langevin equation with an energy function $H(\mathbf{e})$ whose local minima are the named attractor states of the polyvagal and trauma literature (regulated calm, fight, flight, freeze, flow, dissociation). The model is implemented as a real-time instrument: a MIDI controller array maps to the state vector; a Python field server computes energy, gradient, and threshold crossings at 50 Hz; audio output (Ableton Live) and 3D fractal visual output (Mandelbulb, projected onto HoloGauze) are driven by the field state via OSC. We demonstrate the instrument in a recorded session and analyse the resulting state trajectory against circumplex predictions. The model makes predictions that the circumplex cannot: phase transitions into and out of attractor states, the adaptive function of high effective temperature (ADHD modifier), and the depth asymmetry of freeze versus regulated calm.

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1 1. Introduction

1.1 1.1 The Gap in the Literature

The handbook of music and emotion (Juslin & Sloboda, 2010) runs to nearly a thousand pages. The dominant quantitative framework across it is Russell’s (1980) valence–arousal circumplex: emotions as points in a two-dimensional space defined by hedonic valence (pleasant–unpleasant) and arousal (activated–deactivated).

The circumplex is powerful for rating and classification. It is not a dynamical model. It has no energy function, no gradient, no notion of attractor or basin depth, no mechanism for transition between states, no prediction of which transitions are easy and which require large perturbations. It describes a *taxonomy* of emotional positions,

not a *physics* of emotional motion.

Juslin’s BRECVEMA framework (Juslin, 2013) provides a rich taxonomy of the *mechanisms* by which music induces emotion (brainstem reflex, rhythmic entrainment, conditioning, visual imagery, episodic memory, musical expectancy, appraisal). It does not model the *dynamics* that result: how these mechanisms combine to move a listener through state space, how long states persist, what determines escape.

This paper presents a model that does both.

1.2 1.2 The Soma-Field Model

The soma-field model (Johnson, 2026a) defines emotional state as a continuous vector field on the body–brain system. In the musical context, we restrict to a finite-dimensional discretisation: $\mathbf{e}(t) \in \mathbb{R}^{16}$, with 8 emotional modes each carrying a somatic and a cognitive intensity component.

The model imports three structures from mathematical physics via the method of mathematical co-identification (Johnson, 2026b):

1. **Energy function** $H(\mathbf{e})$ from the Hopfield network (Hopfield, 1982; Hertz, Krogh, & Palmer, 1991) — the emotional landscape has local minima (attractor states) and energy barriers between them
2. **Langevin dynamics** — state evolves under gradient descent plus thermal noise; the noise amplitude is the *effective temperature* T_{eff}
3. **Threshold function** — conscious emotional experience arises when field amplitude exceeds a perception threshold θ

1.3 1.3 Why Music

Music is uniquely suited to driving the field. BRECVEMA’s mechanisms act as *forcing functions* on the energy landscape: rhythmic entrainment modulates γ (damping); musical expectancy and resolution create transient wells and barriers; appraisal shifts the bias vector $\mathbf{b}$. The soma-field model provides the dynamical substrate into which BRECVEMA mechanisms plug as parameter modulations.

2 2. The Model

2.1 2.1 State Space

$$\mathbf{e}(t) = (e_1^s, \dots, e_8^s, e_1^c, \dots, e_8^c) \in [0, 1]^{16}$$

where e_i^s is the somatic intensity and e_i^c the cognitive intensity of emotional mode i . The eight modes are: *calm*, *anger/fight*, *anxiety/flight*, *grief*, *freeze/dissociation*, *hypervigilance*, *flow/absorption*, *joy*.

2.2 2.2 Energy Function and Attractors

$$H(\mathbf{e}) = \frac{1}{2} \mathbf{e}^\top W \mathbf{e} - \mathbf{b}^\top \mathbf{e}$$

The coupling matrix W (symmetric, negative semi-definite on the basin of each attractor) encodes which emotional modes co-activate and which compete. The bias vector $\mathbf{b}$ encodes the resting depth of each attractor.

Named attractors match the polyvagal hierarchy and trauma literature: *regulated calm* (global minimum), *fight*, *flight* (shallow saddle), *freeze* (deep isolated minimum), *flow*, *dissociation*.

2.3 2.3 Dynamics

$$\gamma \dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \sqrt{2D} \xi(t)$$

where γ is damping, D is diffusion (noise temperature), and $\xi(t)$ is white noise. The effective temperature is $T_{\text{eff}} = D/\gamma$.

2.4 2.4 Threshold and Conscious Experience

$$\mathcal{P}_i(t) = \mathbf{1}[\max(e_i^s, e_i^c) > \theta]$$

Mode i crosses into conscious experience when its amplitude exceeds threshold θ . Sub-threshold activity is real and causally active but not consciously perceived — matching phenomenological accounts of interoception and pre-verbal affect.

3 3. The Instrument

3.1 3.1 Hardware Architecture

(See *instrument/DESIGN.md* for full specification.)

Layer	Hardware
State input	2× MIDI Fighter Twister (16 encoders each)
Scene control	2× Elgato Stream Deck XL + Bitfocus Companion
Trajectory sequencer	Akai Fire (iSotonik hack)
MIDI routing	Bome MIDI Translator Pro → single virtual port
Audio output	Ableton Live Suite + Max4Live OSC receiver
Visual output	TouchDesigner (Mandelbulb shader) → HDMI → HoloGauze

Layer	Hardware
Field server	Python 3.14, 50 Hz update rate

3.2 3.2 MIDI Mapping

Twister 1 encodes the 8 somatic components; Twister 2 encodes the 8 cognitive components. Each encoder’s turn value maps to $e_i \in [0, 1]$; push toggles mute. Encoders 9–12 on each Twister control field parameters (γ, D, θ) and neurotype modifiers.

3.3 3.3 Audio Rendering

The Max4Live device receives OSC from the Python server and maps:

Field quantity	Audio parameter
$H(\mathbf{e})$	Macro energy — master filter cutoff, reverb size
$\ \nabla H\ $	Rhythmic density / gate rate
T_{eff}	Noise floor, stochastic modulation depth
Threshold crossing $\mathcal{P}_i$	Trigger: note-on for mode i
Nearest attractor	Scene/track selection

3.4 3.4 Visual Rendering

The Mandelbulb power parameter is driven by H ; rotation speed by $\|\nabla H\|$; colour temperature by T_{eff} . Threshold crossings trigger particle bursts. Output via HDMI to a short-throw projector onto HoloGauze screen.

4 4. Demonstration Session

(To be completed — recorded session analysis)

4.1 4.1 Protocol

4.2 4.2 State Trajectory Analysis

4.3 4.3 Comparison with Circumplex Predictions

5 5. Discussion

5.1 5.1 What the Model Adds to BRECVEMA

5.2 5.2 Phase Transitions and Musical Catharsis

5.3 5.3 The ADHD Temperature: A Reframing

The elevated T_{eff} of the ADHD modifier is not purely pathological. Hertz, Krogh, and Palmer (1991) observed that thermal noise in Hopfield networks “makes it possible to kick the system out of spurious local minima” that would trap a deterministic system permanently. In the musical context, a high-temperature listener is not necessarily worse at music engagement — they are harder to trap in a single emotional state, which may be a distinct form of musical sensitivity.

5.4 5.4 Limitations

5.5 5.5 Future Work

- Empirical validation against self-report and psychophysiological measures

- Extension to the full infinite-dimensional field (soma-field-paper §4)
 - Multi-listener coupling (ensemble / therapeutic dyad)
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6 References

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